

Kenar Narayaka

kxnar.github.io | github.com/kxnar | kenar.narayaka@cs.ox.ac.uk | linkedin.com/in/narayaka | +44 7918 448877

EDUCATION

University of Oxford – Christ Church

MCompPhil Computer Science & Philosophy

Oxford, England

2025 – 2029

Relevant Coursework: Probability, Linear Algebra, Discrete Maths, Design & Analysis of Algorithms, Algorithms & Data Structures, Continuous Maths, Functional Programming, Imperative Programming, Logic & Proof, Models of Computation, Philosophical Topics in Logic & Probability, Alan Turing on Computability & Intelligence, General Philosophy, Knowledge & Reality

Loughborough College

A-Level Maths, Further Maths, Computer Science - A*AA

Loughborough, England

2023 – 2025

EXPERIENCE

CapitOx

Jan. 2026 - Apr. 2026

Quant Academy – Oxford-Man Institute | C++, Python, NumPy, SciPy, Matplotlib

- Selected out of 90 people for a competitive programme covering mean-variance portfolio construction, factor models, signal combination, and the design and evaluation of multi-signal systematic strategies
- Built a Monte Carlo pricing pipeline for a European put under a 2-component generalised-normal mixture model, stable log-density evaluation and separating target distribution, proposal mechanism, and estimator logic for controlled comparison.
- Implemented 3 estimators with proposal tuning, convergence diagnostics, and seeded experiments; evaluated across 100+ trials and achieved 5x lower variance and 60x lower runtime at matched error relative to baseline random-walk Metropolis-Hastings.

Hitachi Astemo

Jun. 2024 - Aug. 2024

Backend Engineer Intern – Human Resources | PHP, Fat-Free Framework (F3), Postgres, Docker

- Replaced paper-based HR processes covering ~15,000 employee records with administrative tools, translating legacy forms, approval states, and organisational relationships into normalised Postgres schemas.
- Designed and Dockerised REST APIs and a multi-stage approval engine with server-side validation, RBAC, immutable audit history, and fulfilment tracking; reduced end-to-end request turnaround from ~7 days to 5 hours (-97%)
- Implemented permission-scoped CRUD dashboard and coordinate-based PDF generation connected to network printers, then extracted form components and print-template logic into reusable tooling for later workflows.

PROJECTS

Fracta - 3D Fractal Simulator | JavaScript, three.js, GLSL

- Built a real-time browser-based renderer for fractal families, combining signed-distance estimation with custom GLSL raymarching, configurable lighting and colour pipelines, and camera controls exposed by a web interface.
- Implemented distance-adaptive step sizing that used estimated scene geometry to reduce unnecessary samples and increase frame rate by up to 40%, then added cone-marching for a further ~20% improvement over the original pipeline.
- Profiled performance across camera distances, iteration counts, and fractal params to balance throughput against close-range geometric detail; awarded 69/70, highest OCR Computer Science NEA score in school history.

shotLab - Basketball Jumpshot Analyser | Tauri v2, Rust, ONNX Runtime, FFmpeg

- Built a desktop app that analyses jumpshot form from a single phone video, with every stage of inference running locally so no footage leaves the device; processes a 5-second clip in 50 seconds on my CPU alone.
- Ran top-down RTMPose pose estimation via ONNX Runtime with a SimCC keypoint decoder, and decoded frames in FFmpeg against true presentation timestamps so velocity and timing metrics stay correct on the variable frame rate phone cameras produce.
- Segmented continuous multi-shot recordings into individual reps and their phases, extracting 28 biomechanical parameters across 6 kinematic families; encoded metric availability in Rust's type system so a cue is only surfaced when the keypoints it depends on were actually detected.

TECHNICAL SKILLS

Languages: Haskell, Scala, Python, C++, PHP, JavaScript, SQL, GLSL, Rust

Frameworks & Libraries: NumPy, SciPy, Matplotlib, three.js, Fat-Free Framework, Tauri v2

Tools: Postgres, Git, Linux, tmux, Docker

Concepts: Backend Development, Database Design, Graphics Programming, REST APIs